

Object Oriented Programming (CSEG1044)

Unit 06: Capstone Project Themes

Phase 5: Testing, Build, and Demo

Project: Clinic Appointment System

1. Introduction

Phase 5 is the final stage of the Clinic Appointment System project, where the focus is on ensuring that the system is stable, error-free, and ready for demonstration and evaluation. This phase includes testing core functionalities, verifying system behavior under different conditions, building the final project, and preparing a smooth demo workflow.

2. Goal of This Phase

The primary goal of this phase is to:

- Ensure the system runs without errors
- Validate all core functionalities through testing
- Prepare the project for final build and execution
- Demonstrate a complete working workflow
- Make the system ready for viva and evaluation

3. Priority Test Cases

The following test cases are essential to verify system correctness:

3.1 Prevent Overlapping Appointments

- **Objective:** Ensure that a doctor cannot have two appointments at the same time
- **Test Case:**
 - Book an appointment for Doctor A at 10:00 AM
 - Try booking another appointment for the same doctor at 10:00 AM
- **Expected Result:** System should reject the second booking

3.2 Block Booking Outside Schedule

- **Objective:** Restrict bookings outside the doctor's available time slots
- **Test Case:**
 - Doctor schedule: 9:00 AM – 5:00 PM

- Try booking at 7:00 PM
- **Expected Result:** System should display an error message

3.3 Verify Rescheduling Updates History

- **Objective:** Ensure rescheduling updates records correctly
- **Test Case:**
 - Book appointment at 11:00 AM
 - Reschedule to 2:00 PM
- **Expected Result:**
 - Old appointment is updated or logged
 - New timing is reflected correctly

3.4 Validate Patient Registration Fields

- **Objective:** Ensure correct input validation
- **Test Case:**
 - Leave required fields empty
 - Enter invalid phone number
- **Expected Result:**
 - System should show validation errors

5. Recommended Demo Flow

A proper demo should follow a logical sequence:

1. Add doctor schedule
2. Register a new patient
3. Book an appointment
4. Attempt conflicting booking
5. Show system rejecting invalid booking

This flow demonstrates both functionality and validation.

8. Conclusion

Phase 5 ensures that the Clinic Appointment System is reliable, functional, and ready for final presentation. Proper testing, clean build, and a well-prepared demo are crucial for successful evaluation. By focusing on core features and avoiding common mistakes, the project can effectively demonstrate real-world application of Object-Oriented Programming concepts.